



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 27/06/2019*

PROBLEM

We observe two samples of a realization of the discrete-time random process $X[n] = S[n] + W[n] = A \cos(2\pi F_0 n) + W[n]$, $n=0,1$, where A is the amplitude and F_0 the frequency of the signal of interest (SOI), they are modelled as unknown deterministic parameters. We know that $A \neq 0$ and $0 < F_0 < 1/2$. $\{W[n]; n=0,1\}$ are independent random variables modeling additive noise; each noise sample is Gaussian distributed, with zero mean and variance σ_w^2 .

(1) Derive the joint probability density function (pdf) $f_{\mathbf{x}}(\mathbf{x}; A, F_0)$ of the observed data vector $\mathbf{X} \triangleq [X[0] \ X[1]]^T = \mathbf{S} + \mathbf{W}$ and the log-likelihood function $\ln L(A, F_0) = \ln f_{\mathbf{x}}(\mathbf{x}; A, F_0)$. Derive the Signal to Noise Ratio (SNR) defined as $\gamma \triangleq E\{\|\mathbf{S}\|^2\} / E\{\|\mathbf{W}\|^2\}$.

(2) Derive the joint Maximum Likelihood (ML) estimator of A and F_0 given the observed data vector $\mathbf{X} \triangleq [X[0] \ X[1]]^T$.

(3) Derive the bias and the Mean Square Error (MSE) of the ML estimator of the amplitude A .

(4) Derive the elements of the 2x2 Fisher Information Matrix (FIM) and then derive the Cramér-Rao lower bound (CRB) on the estimate of A and establish whether the ML estimator of A is efficient or not.

(5) Describe what happens when $\gamma \rightarrow \infty$ and show that the ML estimators of A and F_0 are consistent.



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SOLUTION

(1) Probability density function $f_{\mathbf{x}}(\mathbf{x}; A, F_0)$ and $\ln L(A, F_0)$

$$X[0] = A + W[0] \in N(A, \sigma_w^2), \quad X[1] = A \cos(2\pi F_0) + W[1] \in N(A \cos(2\pi F_0), \sigma_w^2)$$

$$f_{\mathbf{x}}(\mathbf{x}; A, F_0) = f_{x_0}(x_0; A, F_0) f_{x_1}(x_1; A, F_0) = \frac{1}{2\pi\sigma_w^2} \exp \left[-\frac{(x_0 - A)^2 + (x_1 - A \cos(2\pi F_0))^2}{2\sigma_w^2} \right]$$

$$\ln L(A, F_0) = \ln f_{\mathbf{x}}(\mathbf{x}; A, F_0) = -\ln(2\pi\sigma_w^2) - \frac{(x_0 - A)^2 + (x_1 - A \cos(2\pi F_0))^2}{2\sigma_w^2}$$

$$\gamma = \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}} = \frac{E\{A^2 + A^2 \cos^2(2\pi F_0)\}}{E\{W^2[0] + W^2[1]\}} = \frac{A^2(1 + \cos^2(2\pi F_0))}{2\sigma_w^2}$$

(2) Maximum Likelihood (ML) estimator of A and F_0

$$\frac{\partial \ln L(A, F_0)}{\partial F_0} = -\frac{1}{2\sigma_w^2} [4\pi A \sin(2\pi F_0)(x_1 - A \cos(2\pi F_0))] = -\frac{2\pi A \sin(2\pi F_0)}{\sigma_w^2} (x_1 - A \cos(2\pi F_0))$$

$$\frac{\partial \ln L(A, F_0)}{\partial F_0} = 0 \rightarrow x_1 - A \cos(2\pi F_0) = 0 \rightarrow \cos(2\pi F_0) = \frac{x_1}{A}$$

$$F_0 = \frac{1}{2\pi} \arccos\left(\frac{x_1}{A}\right)$$

$$\begin{aligned} \frac{\partial \ln L(A, F_0)}{\partial A} &= -\frac{1}{2\sigma_w^2} [-2(x_0 - A) - 2\cos(2\pi F_0)(x_1 - A \cos(2\pi F_0))] \\ &= \frac{1}{\sigma_w^2} [x_0 - A + \cos(2\pi F_0)x_1 - A \cos^2(2\pi F_0)] \\ &= -\frac{1 + \cos^2(2\pi F_0)}{\sigma_w^2} \left[A - \frac{x_0 + \cos(2\pi F_0)x_1}{1 + \cos^2(2\pi F_0)} \right] \end{aligned}$$

$$\frac{\partial \ln L(A, F_0)}{\partial A} = 0 \rightarrow [x_0 - A + \cos(2\pi F_0)x_1 - A \cos^2(2\pi F_0)] = 0$$

Recalling that $x_1 - A \cos(2\pi F_0) = 0 \rightarrow x_1 = A \cos(2\pi F_0) \rightarrow \cos(2\pi F_0)x_1 = A \cos^2(2\pi F_0)$,



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we get:

$$x_0 - A + \cos(2\pi F_0)x_1 - A \cos^2(2\pi F_0) = x_0 - A + A \cos^2(2\pi F_0) - A \cos^2(2\pi F_0) = x_0 - A = 0$$

$$\rightarrow A = x_0 \rightarrow \hat{A}_{ML} = x_0$$

Replacing A with $\hat{A}_{ML} = x_0$ in the first equation we get the ML estimate of the frequency:

$$F_0 = \frac{1}{2\pi} \arccos\left(\frac{x_1}{A}\right) \rightarrow \hat{F}_{0ML} = \frac{1}{2\pi} \arccos\left(\frac{x_1}{x_0}\right)$$

(3) Bias and the Mean Square Error (MSE) of the ML estimator of the amplitude A .

$$\hat{A}_{ML} = x_0 \rightarrow E\{\hat{A}_{ML}\} = E\{x_0\} = A \rightarrow \hat{A}_{ML} \text{ is unbiased.}$$

$$\text{MSE}\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = \sigma_w^2$$

(4) Cramér-Rao lower bounds (CRBs) on the estimate of A and F_0

$$\frac{\partial \ln L(A, F_0)}{\partial A} = -\frac{1 + \cos^2(2\pi F_0)}{\sigma_w^2} \left[A - \frac{x_0 + \cos(2\pi F_0)x_1}{1 + \cos^2(2\pi F_0)} \right]$$

$$\frac{\partial^2 \ln L(A, F_0)}{\partial A^2} = -\frac{1 + \cos^2(2\pi F_0)}{\sigma_w^2} \rightarrow I_{11} = -E\left\{\frac{\partial^2 \ln L(A, F_0)}{\partial A^2}\right\} = \frac{1 + \cos^2(2\pi F_0)}{\sigma_w^2}$$

$$\frac{\partial \ln L(A, F_0)}{\partial F_0} = -\frac{2\pi A \sin(2\pi F_0)}{\sigma_w^2} [x_1 - A \cos(2\pi F_0)]$$

$$\frac{\partial^2 \ln L(A, F_0)}{\partial F_0^2} = -\frac{4\pi^2 A}{\sigma_w^2} [\cos(2\pi F_0)(x_1 - A \cos(2\pi F_0)) + A \sin^2(2\pi F_0)]$$

$$I_{22} = -E\left\{\frac{\partial^2 \ln L(A, F_0)}{\partial F_0^2}\right\} = \frac{4\pi^2 A^2 \sin^2(2\pi F_0)}{\sigma_w^2}$$



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$$\frac{\partial \ln L(A, F_0)}{\partial F_0} = -\frac{2\pi A \sin(2\pi F_0)}{\sigma_w^2} (x_1 - A \cos(2\pi F_0)) = -\frac{2\pi \sin(2\pi F_0) x_1}{\sigma_w^2} A + \frac{2\pi \sin(2\pi F_0) \cos(2\pi F_0)}{\sigma_w^2} A^2$$

$$\begin{aligned} \frac{\partial^2 \ln L(A, F_0)}{\partial A \partial F_0} &= \frac{\partial}{\partial A} \left[-\frac{2\pi \sin(2\pi F_0) x_1}{\sigma_w^2} A + \frac{2\pi \sin(2\pi F_0) \cos(2\pi F_0)}{\sigma_w^2} A^2 \right] \\ &= -\frac{2\pi \sin(2\pi F_0) x_1}{\sigma_w^2} + \frac{4\pi A \sin(2\pi F_0) \cos(2\pi F_0)}{\sigma_w^2} \end{aligned}$$

$$I_{12} = -E \left\{ \frac{\partial^2 \ln L(A, F_0)}{\partial F_0^2} \right\} = -\frac{2\pi A \sin(2\pi F_0) \cos(2\pi F_0)}{\sigma_w^2}$$

$$\begin{aligned} CRB(A) &= \frac{I_{22}}{I_{11}I_{22} - I_{12}I_{21}} = \frac{1}{I_{11} - \frac{(I_{12})^2}{I_{22}}} \\ &= \frac{1}{\frac{1 + \cos^2(2\pi F_0)}{\sigma_w^2} - \frac{4\pi^2 A^2 \sin^2(2\pi F_0) \cos^2(2\pi F_0)}{\sigma_w^4} \cdot \frac{\sigma_w^2}{4\pi^2 A^2 \sin^2(2\pi F_0)}} \\ &= \frac{\sigma_w^2}{1 + \cos^2(2\pi F_0) - \cos^2(2\pi F_0)} = \sigma_w^2 \end{aligned}$$

$$CRB(A) = \sigma_w^2 = \text{MSE} \{ \hat{A}_{ML} \} \rightarrow \hat{A}_{ML} \text{ is efficient.}$$

(5) When $\gamma \rightarrow \infty$ we have that $X[0] \rightarrow A$, $X[1] \rightarrow A \cos(2\pi F_0)$, so $\hat{A}_{ML} = x_0 \rightarrow A$ and

$$\hat{F}_{0ML} = \frac{1}{2\pi} \arccos \left(\frac{x_1}{x_0} \right) \rightarrow \frac{1}{2\pi} \arccos \left(\frac{A \cos(2\pi F_0)}{A} \right) = F_0. \text{ Hence, the two estimators are consistent.}$$